PaperLess Campus

Campus approvals, signed in one tap and verifiable by anyone.

Team

budYweisers

College

VIT Pune

Members

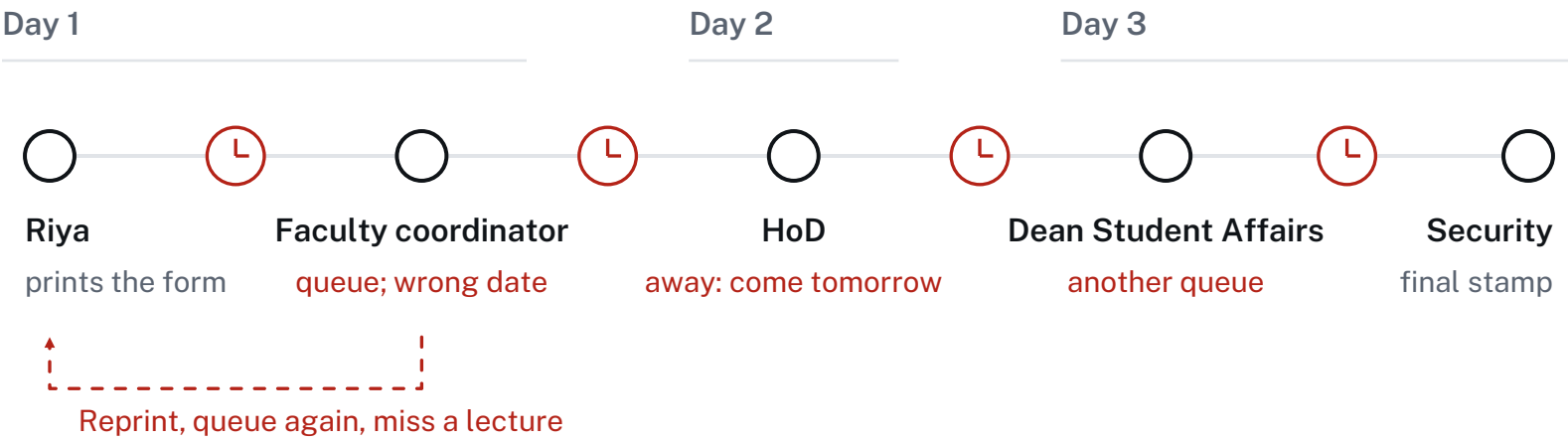
Shruti Dwivedi

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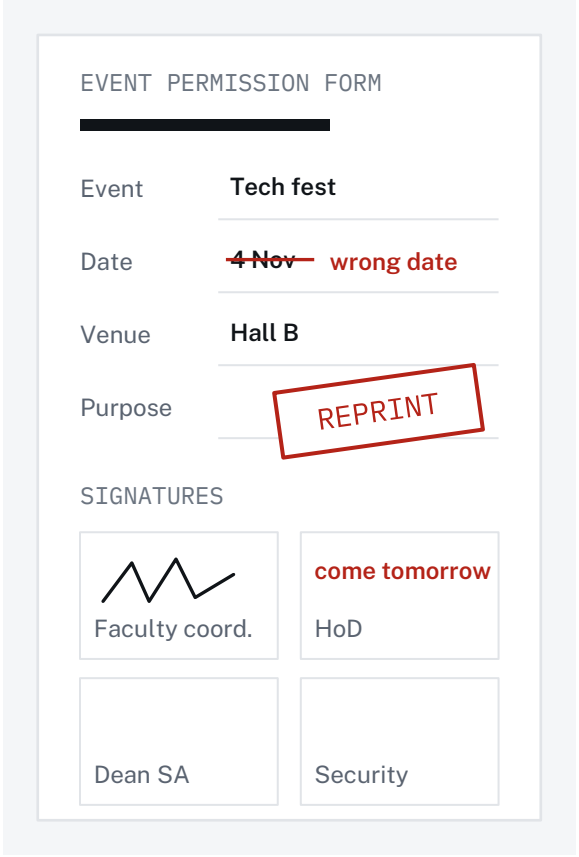
Problem Statement

One approval, four cabins, three days: Riya's permission letter for her tech fest.



4 signatures chased **3** days **2** lectures missed **0** status updates

Many students chase **~15 signatures a month**; it varies from college to college.



The paper form Riya carries

Research references: Riya's journey: a typical event-permission chain on our campus. ~15 signatures a month: team estimate that varies by college, to be measured in a 50-student survey.

Scale of the Problem

Manual approvals cost everyone time and leave institutions without an audit trail.

4.50 crore

students in Indian higher education

1,289 universities · 48,246 colleges

~67 crore

paper approvals a year (est.)

~15 forms per student; ~200 crore sheets

May 2027

DPDP Rules' core duties apply

consent, retention and breach records

Who carries the manual load today

Students

Queue in class hours, with no live status

Faculty and HoDs

Interrupted to sign near-identical slips

Admin office

Physical files; no audit trail for NAAC or DPDP

Accessibility

Remote and differently-abled students must reach every cabin

India already has the digital rails. The everyday campus approval is the missing last mile.

Research references: AISHE 2023-24, Ministry of Education (released Jul 2026). DPDP Rules 2025 (MeitY, notified Nov 2025; core duties from May 2027). Est.: 15 forms × 3 sheets per student a year.

Proposed Solution

A student submits once; the system routes it by role, collects signatures and seals the result.

Request
You fill the form

Route
By role, not by name

Sign
One tap, OTP or eSign

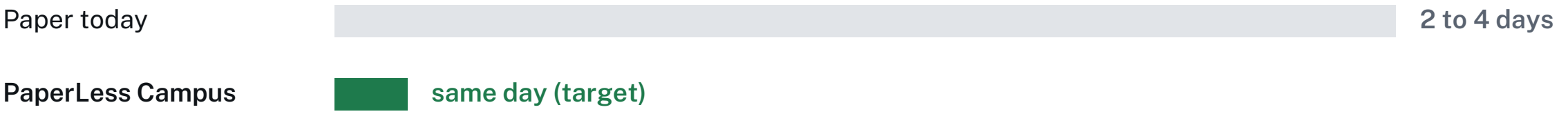
Seal
Hash on chain

Verify
Scan code on the PDF

Works for

Event permission · On-duty · Bonafide · Internship NOC · No-dues · LOR · Scholarship

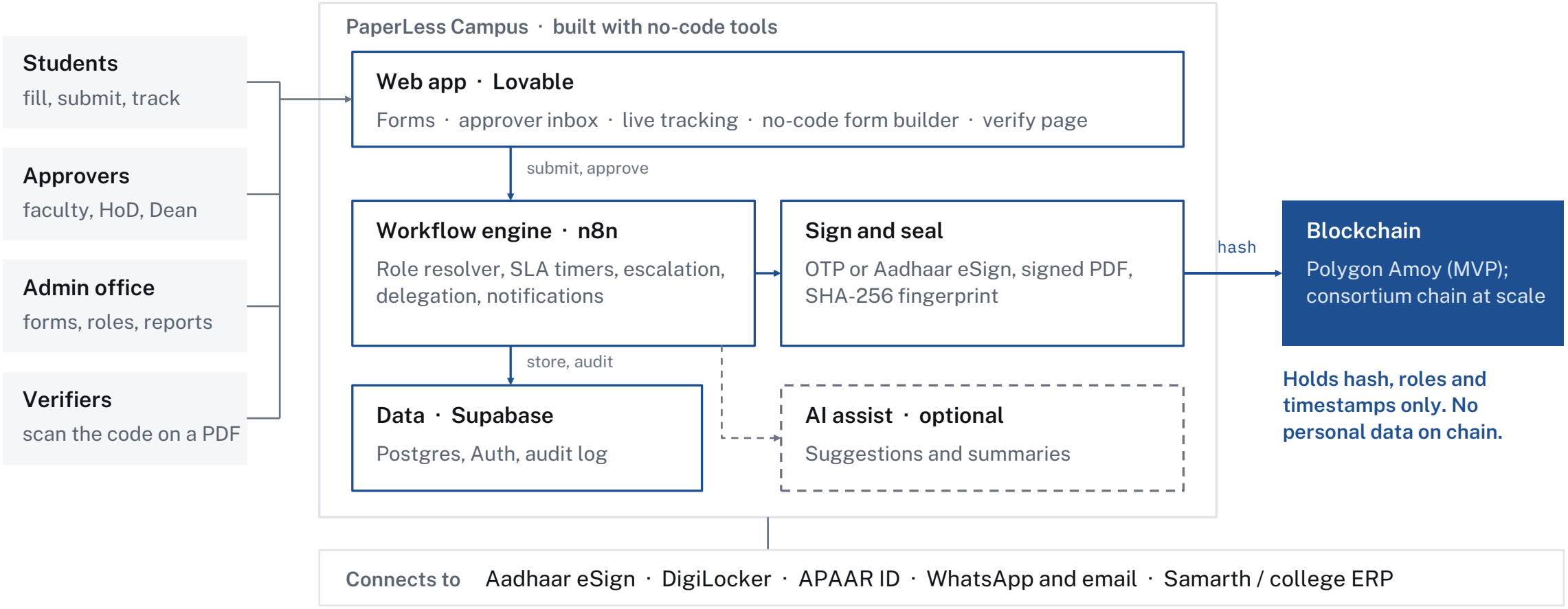
Time to a signed approval



Research references: Paper turnaround is a typical range on our campus, to be measured in the pilot; same day is the pilot target.

System Architecture

Every request flows through one no-code platform; only its fingerprint goes on chain.



Tech Stack

Every layer is no-code or AI-prompted, so a college IT cell can run it without developers.

Layer	Primary tool	Backup	Role in PaperLess Campus	Type
Web app	**Lovable**	Softr	Student, approver, admin and verify screens	**No-code**
Workflow and routing	**n8n**	Make	Role routing, SLA timers, escalation, delegation	**No-code**
Data, login, files	**Supabase**	Airtable	Postgres, Auth, Storage; row-level security	**Managed**
AI assist (optional)	**Gemini or Claude in n8n**	OpenAI API	Completeness check, summaries, suggestions	**AI**
Indian languages	**Sarvam AI**	Gemini	Status chat in Hindi and Marathi	**AI**
E-signature	**OTP sign; Aadhaar eSign**	DSC tokens	Aadhaar eSign via Leegality or Digio	**Legal**
Blockchain	**Polygon Amoy testnet**	Hyperledger Besu	Document hash; consortium chain at scale	**Trust**
PDF and alerts	**Google Docs via n8n**	PDFMonkey	Signed PDF; email and WhatsApp alerts	**No-code**

Free tiers cover the pilot. Open-source n8n and Postgres avoid lock-in. Shaded rows are optional AI help.

Workflow and AI Assistance

The system knows who must sign, in order, and helps them decide in seconds.

Event permission, routed by role

- **Student submits**
  Fills the form, adds files
- **Faculty coordinator**
  SLA 48 h, then a nudge
- **HoD**
  On leave? Goes to the delegate
- **Security**
  in parallel
- **Dean Student Affairs**
  Missed SLA? Auto-escalates
- **Approved and sealed**

Optional AI help

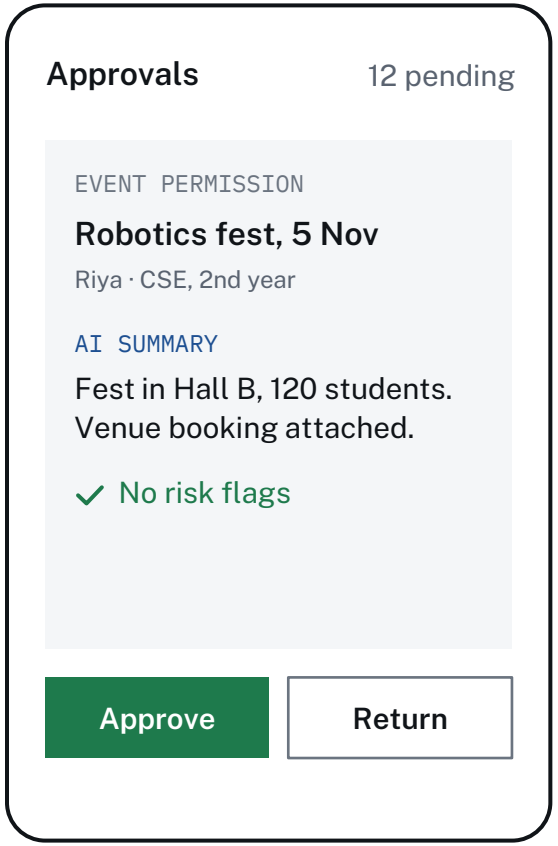
Field suggestions
Suggests a missing detail

Completeness check
Catches a missing attachment early

Two-line summary
Approver reads two lines, not pages

Status chat
"Where is my NOC?" in English, Hindi, Marathi

People decide; AI only suggests.



Concept screen: HoD inbox

Trust and Verification

Change one character and the proof breaks.

Signed PDF → **SHA-256 fingerprint** → **Written on chain** → **Verify code on the PDF**

Approved by Guide, T&P officer, HoD

85dcad7e…f4838234

Hash, signer roles, timestamps

Opens the public verify page

Document a verifier holds	Re-computed hash		Hash on chain	
Original document	85dcad7e…f4838234	=	85dcad7e…f4838234	Verified
Request no. 0412 edited to 0413	3917b63d…e973ddc4	≠	85dcad7e…f4838234	Tampered

IT Act 2000, s.5
e-sign as valid as wet ink

Aadhaar eSign
notified in the Second Schedule

BSA 2023, s.61
electronic records admissible

DPDP Rules 2025
no personal data on chain

Same model as
India's National Blockchain Framework

Research references: Hashes are real SHA-256 values of a sample record. Legal: IT Act 2000 s.3A, s.5, Second Schedule; Bharatiya Sakshya Adhiniyam 2023 s.61; DPDP Rules 2025 (MeitY, Nov 2025).

Impact and Benefits

One campus saves lakhs of sheets; India saves tens of thousands of tonnes of CO2e.

Per year, estimated (one university = 10,000 students)	One university	All of India
Approvals moved off paper	**1.5 lakh**	**~67 crore**
Sheets of paper not printed	**~4.5 lakh**	**~200 crore**
CO2e avoided	**≥5.7 t**	**≥25,400 t**
Trees not cut	**~54**	**~2.4 lakh**
Student-hours returned	**~1.1 lakh**	not estimated
Faculty-hours returned	**~8,000**	not estimated

UN Sustainable Development Goals

SDG 4 Quality Education
class time, not queue time

SDG 9 Industry and Infrastructure
shared digital rails

SDG 12 Responsible Consumption
single-use paper removed

SDG 13 Climate Action
live CO2-saved counter

SDG 15 Life on Land
trees not cut

SDG 16 Strong Institutions
every decision traceable

Research references: Assumptions: 15 forms × 3 sheets per student; 45 min chasing per form; 2 signers, 2 min paper vs 20 s digital. Per sheet: 12.7 g CO2e, 1/8,333 tree (Environmental Paper Network calculator v3.2, via PaperCut).

Roadmap and Future Scope

From one department to every institution that runs on signatures.

MVP prototype
Submit, route, sign, seal, verify

Gate: Admin approves

Department pilot
One department, one semester

Gate: Turnaround halved

University rollout
Aadhaar eSign and DigiLocker

Gate: 3 partner colleges

College network
Partner colleges as chain nodes

Same engine beyond education: roles, a sign-off chain and proof

- Government offices
- Hospitals
- Banks and NBFCs
- Corporates
- Municipal and land
- Schools and boards

₹50
per student a year: about what a campus spends printing these forms (est.)

NAAC 6.2.3
accreditation already counts e-governance in administration

19.21 lakh
government files weeded out in one month (Special Campaign 5.0)

Research references: Est.: 4.5 lakh sheets at ₹1-2 a page = ₹4.5-9 lakh vs ₹5 lakh at ₹50 × 10,000 students. NAAC metric 6.2.3. Special Campaign 5.0: PIB, Oct 2025.